

Calibrate Once, Measure Within Support: Multicalibration for Unbiased Prevalence Estimation from Political Text

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Abstract

Political scientists increasingly use classifiers, and now large language models (LLMs), to estimate how prevalent concepts are in natural language, such as topics in legislative agendas or categories in open-ended survey responses. Standard validation (reporting accuracy or AUC on a labeled sample) certifies neither valid prevalence estimates nor their cross-context comparability. When the target differs from the validation sample (covariate shift), an accurate classifier can still produce badly biased prevalence estimates, and the usual corrections must be redone for every new target. We show that *multicalibration*, calibration conditional on input features rather than on average, is sufficient for unbiased prevalence estimation under any covariate shift along the calibrated features, yielding an instrument calibrated once and applied to any target within its support. A cross-national LLM topic-coding application shows near-zero bias under shift, where standard methods miss by a large margin, and the correction works on the model’s bare binary outputs without eliciting probabilities. A hand-verifiable simulation and an exact-ground-truth benchmark on a non-text classifier, both in the appendix, corroborate the mechanism.

Introduction

A growing share of empirical political science depends on machines reading text. Researchers use automated classifiers, and increasingly large language models (LLMs), for example, to code policy agendas across countries (Baumgartner et al. 2006; Benoit et al. 2026), detect protest and conflict events (Overos et al. 2024; King and Lowe 2003), score democratic institutions (Weidmann et al. 2026), categorize open-ended survey responses (Mellon et al. 2024), and label social media at scale (Gilardi et al. 2023; Ornstein et al. 2025). The quantity of interest is usually not the label on any single document but an *aggregate*: what fraction of a parliament’s questions concern law and crime, or how often a topic appears in one country’s media versus another’s. These are *prevalence* (or *quantification*) problems (González et al. 2017; Hopkins and King 2010).

Text-as-data has a long lineage, from word-scaling of policy positions (Laver et al. 2003) and automated event extraction (King and Lowe 2003) to the supervised methods synthesized by Grimmer et al. (2022) and Grimmer and Stewart (2013). Throughout, researchers validate a classifier by how well it agrees with human labels. They report that agreement as accuracy, F_1 , or AUC (Halterman and Keith 2025), and judge it against the intercoder reliability of the human benchmark itself (Hayes and Krippendorff 2007). A classifier that clears a high bar is then applied to new corpora, other countries, earlier decades, different document types (Benoit et al. 2026). This rests on an unexamined assumption: that performance on the validation sample transfers to the populations where the instrument is used. For the prevalence estimands these studies target, that assumption is frequently wrong. Agreement on a validation corpus is necessary but not sufficient.

Separating positives from negatives is not the same as counting them. A classifier can rank documents almost perfectly and still misstate the *proportion* that are positive. How far off it is depends on the population it is applied to. If a new corpus contains more of the cases the classifier tends to misjudge, its errors no longer cancel, even when the meaning of the category has not changed. This is *covariate shift*. Across corpora the mix of documents changes (languages, venues, eras, authors), but the rule linking content to category stays fixed. Classical quantifiers assume the opposite. Under *label shift*, documents of a category look the same everywhere and only their overall frequency changes, which rarely holds across languages and venues. A high AUC is no safeguard against this, and standard validation does not reveal it (Adcock and Collier 2001).

Existing tools each address part of the problem, but none yields a *reusable* instrument. Quantification methods estimate prevalence from imperfect classifiers (González et al. 2017). Classify & Count, its bias-corrected relatives (Adjusted Count / Rogan-Gladen (Rogan and Gladen 1978), the Saerens-Latinne-Decaestecker EM algorithm (Saerens et al. 2002), distribution-matching (Maletzke et al. 2018)), and the classifier-free ReadMe family (Hopkins and King 2010; Jerzak et al. 2023) all assume label shift, and so do not transport under covariate shift. Design-based supervised learning (Egami et al. 2023) and prediction-powered inference (Angelopoulos et al. 2023) debias an estimate using gold labels drawn from the target itself. Both are target-specific, because each new population needs its own sample. Importance weighting (Shimodaira 2000; Sugiyama et al. 2007) applies under covariate shift, but its density ratio must be re-estimated for each target. A related literature on learned proxies (Knox et al. 2022) and on the reliability of LLM annotations (Bisbee et al. 2024; Argyle et al. 2023; Rodriguez and Spirling 2022) raises the same worry that imperfect measures distort inference. Weidmann et al. (2026) in particular find that LLM classifiers deviate from experts in systematic, feature-dependent directions, which conditioning on features can correct.

What ties a classifier to prevalence is calibration. The simplest estimator just averages the classifier’s scores, and that average is unbiased precisely when the classifier is calibrated, that is, when its predicted probabilities match empirical frequencies in the population at hand. Calibration, however, is a property of one population and need not survive a shift. Empirically it degrades under language and task shift even when accuracy is preserved (Yang et al. 2023; Ren et al. 2025). Confidence elicitation (Tian et al. 2023; Kadavath et al. 2022; Wang et al. 2023) and post-hoc recalibration (Wu and Resnick 2024) aim only at this global calibration, and are fragile in the same way. We show that *multicalibration*, calibration conditional on the input features rather than on average (Hébert-Johnson et al. 2018; Detommaso et al. 2024), fills this gap. A multicalibrated classifier is stable under reweighting of those features. Calibrated once on a labeled source sample, it returns unbiased prevalence estimates on any target whose shift the features capture and that lies within their support (Kim et al. 2022), with no target-specific re-estimation. We develop this into a measurement protocol for political text and show which correction suits which shift regime.

Prevalence Bias Reduction through Multicalibration

Consider a binary outcome $Y \in \{0, 1\}$, features X , and a device $h(X) \in [0, 1]$. The estimand is the prevalence $\pi^* = P^*(Y = 1)$ in a target population, estimated from unlabeled target data $\{X_i\}$ and h . We assume *covariate shift*: $P(X)$ differs between calibration and target while $P(Y | X)$ is stable. Substantively, the coding rule stays fixed. A document with given content receives the same label whatever corpus it comes from, which is what a codebook is meant to guarantee. What changes is the mix of documents. This is the complement of the *label-shift* assumption, which instead holds $P(X | Y)$ fixed while the prior $P(Y)$ moves.¹ Under concept drift, where $P(Y | X)$ itself changes across contexts, no purely statistical method recovers the target prevalence without labels from the target.

A device is *globally calibrated* if $\mathbb{E}[Y | h(X) = p] = p$ for all p . Then $\mathbb{E}[h(X)] = \pi$, and averaging its predictions is unbiased. But calibration on one population need not hold on another. Partition the population into feature-defined groups G with weights w_G , and let $\epsilon_G = \mathbb{E}[h(X) - Y | X \in G]$ be the device’s mean signed error in G . Global calibration requires only $\sum_G w_G \epsilon_G = 0$. Under covariate shift the weights become w_G^* , and the prevalence bias is

$$\hat{\pi}^* - \pi^* = \sum_G w_G^* \epsilon_G,$$

nonzero in general unless every $\epsilon_G = 0$.² A device can be perfectly calibrated on the validation sample yet badly biased on a target that reweights those groups. A high AUC offers no protection. AUC depends only on the ranking of scores, and is therefore invariant to the group errors ϵ_G that produce the bias.

The decomposition explains why existing corrections fail. Output-based corrections (Classify & Count, Rogan-Gladen, global isotonic recalibration (Wu and Resnick 2024)) estimate aggregate positive and error rates on the source. These rates are weighted averages over groups, and they no longer cancel once the target reweights the groups. Label-shift quantifiers (SLD/EMQ (Saerens et al. 2002), distribution-matching, the classifier-free ReadMe family (Hopkins and King 2010; Jerzak et al. 2023)) fail for a different reason. They carry no per-document ϵ_G , but the identity they solve presumes $P(X | Y)$ is stable.

The fix is to drive *every* ϵ_G to zero, not merely their weighted sum. That is what *multicalibration* does. A predictor f is multicalibrated with respect to a class of groups $\mathcal{G}$ if $\mathbb{E}[Y | f(X) = v, X \in G] = v$ for all $G \in \mathcal{G}$ and values v (Hébert-Johnson et al. 2018).³ A multicalibrated predictor then satisfies the Kim et al. (2022) “universal adaptability” guarantee. For any target whose density ratio dP^*/dP is measurable with respect to $\mathcal{G}$,

$$\mathbb{E}^*[f(X)] = \mathbb{E}\left[\frac{dP^*}{dP}(X) f(X)\right] = \mathbb{E}\left[\frac{dP^*}{dP}(X) Y\right] = \mathbb{E}^*[Y] = \pi^*,$$

because the density ratio is constant within each $\mathcal{G}$ -cell, so the within-cell $\epsilon_G = 0$ carries through. One instrument, calibrated once, is unbiased on every target whose shift lies along the calibrated features and within support, with no knowledge of the shift and no target-specific estimation.

¹The heuristic that stable $P(Y | X)$ is natural when features cause the outcome, and stable $P(X | Y)$ when the outcome causes the features, traces to @scholkopf2012causal; we rely on the invariance rather than on adjudicating the causal direction.

²Because ϵ_G is estimated on the source, the identity is exact when each group is homogeneous enough that the density ratio is constant within it, and a first-order approximation otherwise; see SI Section S1.

³A slightly weaker property, *multi-accuracy* (zero average error within each group), already suffices for unbiased prevalence; multicalibration is stronger but the better practical target (SI Section S1.8).

Importance weighting uses the same density ratio, reweighting the labeled source to match the target (Shimodaira 2000; Sugiyama et al. 2007). It is valid under the same assumption, but it corrects the *sample* rather than the *predictor*. The density ratio must therefore be re-estimated for each target, and the method breaks under positivity violations. Multicalibration instead folds the correction into the predictor once, and so matches per-target importance weighting without per-target work.

The guarantee rests on two conditions. First, *support*: the target’s feature values must lie within the calibration support. Second, *expressiveness*: the feature class $\mathcal{G}$ must capture every dimension along which the shift interacts with the outcome, an ignorability condition discharged once, at calibration time. In text the expressiveness condition is the more fragile of the two. Coarse metadata cells can hide latent content that drives Y . To guard against this, we enrich $\mathcal{G}$ with text-derived features such as document embeddings (Rodriguez and Spirling 2022). When either condition fails, estimates degrade, as our out-of-support results show. We obtain the multicalibrated predictor with MCGrad (Tax et al. 2026), a post-hoc gradient-boosting multicalibrator.

Example: LLM coding of political text under covariate shift

We demonstrate the mechanism outlined above with an example text-as-data application: one coding instrument applied across countries and languages.⁴ We use Claude Opus 4.6 to classify 30,000 Comparative Agendas Project texts (Baumgartner et al. 2006) for whether they concern Law & Crime (CAP major topic 12), scored against the published CAP expert codes. The documents span six sub-populations across four countries and languages, 5,000 each: Danish parliamentary questions, Spanish oral questions, U.S. congressional bills, Belgian newspaper articles, Spanish media articles, and Belgian TV news. The design reproduces the researcher’s situation. We code a concept once, then estimate its prevalence across corpora that differ in language, venue, and format. This spread of corpora lets us trace bias across the shift gradient.

We compare two output modes spanning current practice: the discrete Yes/No classification used in essentially all applied studies, and a probability score from direct elicitation of $P(\text{Yes})$, the confidence-elicitation recommendation (Tian et al. 2023; Kadavath et al. 2022). Discrimination is strong in both (AUC 0.960 binary, 0.987 scores; per-language 0.983–0.994). We draw the calibration set ($n \approx 13,400$) equally from four sub-populations, and hold out two others (Spanish media, Belgian TV) as out-of-support targets. These share country and language with the calibration data but add an unseen document type. We calibrate MCGrad on observable metadata only: country, document type, party, decade, and document length. These features serve as the group class $\mathcal{G}$. For the binary condition, MCGrad receives the Yes/No label as a categorical feature, with scores initialized at the calibration base rate, and never sees a probability. For the score condition, it receives $P(\text{Yes})$. We benchmark against Classify & Count, Rogan-Gladen, isotonic regression, importance weighting (IPW), and SLD (SI Table S2).

⁴Two further analyses in the SI support the same mechanism from other angles. A simulation with a hand-verifiable classifier isolates it in a case checkable by hand (SI Section S4), and an American Community Survey employment benchmark with exact ground truth shows the correction is not specific to text or to language models but holds for an ordinary classifier (SI Section S3).

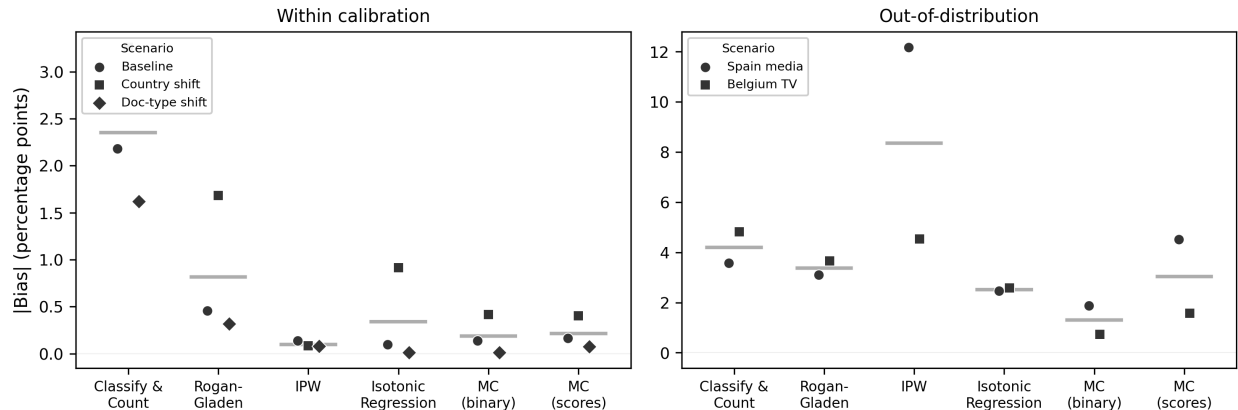


Figure 1. Absolute prevalence bias (percentage points) across the shift gradient: a no-shift baseline, two within-support shifts (country and document-type composition), and two out-of-support targets. Marker shape denotes scenario; horizontal lines are scenario means. The two multicalibration variants (MC binary on Yes/No labels, MC scores on elicited probabilities) are indistinguishable within support (both $\leq 0.4pp$, below the 0.4–0.6pp sampling error in the gold base rate). Full numbers in SI Table S2.

The failure pattern is clear (Figure 1). Classify & Count, the applied default, is biased by +1.6 to +4.8pp across scenarios. Rogan-Gladen reduces this within support but still misses by +3.1 to +3.7pp out of support. Isotonic regression is good within support ($\leq 0.9pp$) but worse out of it (2.5–2.6pp). SLD, the label-shift quantifier, is already biased at baseline (+7.4pp, since the elicited scores are not the calibrated posteriors its EM step assumes) and reaches +21.6pp out of support. This confirms that the label-shift family does not transport under covariate shift.⁵

Multicalibration on the bare Yes/No labels achieves near-zero bias within calibration ($\leq 0.4pp$) and degrades gracefully out of support (+0.7pp on Belgian TV, $-1.9pp$ on Spanish media), from the label and metadata alone. On probability scores it matches this within support ($\leq 0.4pp$) and is somewhat worse out of support ($-4.5pp$ on Spanish media, +1.6pp on Belgian TV). That bare labels match or beat elicited probabilities is the practically important finding. The correction comes from the metadata features and the calibration sample, not from the quality of the model’s confidences. IPW is near-zero within support but fails out of support ($-12.2pp$ on Spanish media), where the target contains feature values absent from the source (a positivity violation). It must also re-estimate density ratios for each target.

Two caveats. First, the gold standard is itself the CAP expert coding. With 5,000 documents per sub-population, the binomial sampling error on a base rate is about ± 0.4 – $0.6pp$, so within-support differences below this should be read as ties. The substantive contrast is between the near-zero-bias methods and the multi-point biases of Classify & Count and SLD. Second, an open-weight replication with Llama 3.3 70B (SI Section S2) reproduces the pattern (MCGrad $\leq 0.2pp$ within calibration), so the findings are not specific to a proprietary model.

⁵ReadMe [hopkinsking2010nonparametric], the canonical political-science quantifier, is classifier-free (it estimates the proportion from raw text rather than correcting the LLM’s outputs), and its bias runs off this scale; we report it in SI Section S7 (Table S4). It fails as the label-shift diagnosis predicts, near-zero only without shift and exceeding +60pp on some shifts; ReadMe2 [jerkzakkinstrezhnev2023improved] shares the assumption and cannot escape it.

Discussion

Prevalence estimation across populations is among the most common uses of text-as-data, and more fragile than current practice acknowledges. Discriminative validation shows only that a classifier separates classes. It is silent on whether the resulting prevalence estimates transport. What makes them transport is calibration conditional on features, which we obtain via multicalibration. A validated classifier then becomes an instrument that is calibrated once and applied to many populations. Our result connects two literatures that developed apart: quantification, which corrected aggregate error rates but not feature-conditional calibration, and multicalibration, which studied individual-level prediction and fairness but not population-level inference.

The payoff is target-independence. Importance weighting, design-based supervised learning, and prediction-powered inference all redo work for each target. A multicalibrated instrument is calibrated once and deployed without target-specific estimation, which is what comparative and over-time research most needs. The choice among methods turns on where labeled data comes from. If a gold sample can be drawn from each target, design-based methods (Egami et al. 2023; Angelopoulos et al. 2023) are efficient and assumption-light, and we recommend them. To measure across many targets without per-target hand-coding, multicalibration provides one reusable instrument, at the cost of the ignorability and support assumptions. ReadMe-family quantifiers (Hopkins and King 2010; Jerzak et al. 2023) are right under label shift but fail under covariate shift (SI Section S7).

The guarantee has limits. It holds only along calibrated dimensions, so calibration data must span the anticipated variation. It also requires labeled calibration data. In zero-shot settings that reintroduces some hand-coding, though far less than per-document labeling and only once. When per-target gold labels are available, design-based methods are the better tool. Under mixtures, the covariate-shift condition holds only approximately. Under concept drift, no purely statistical correction substitutes for new labels. Our applications have measurable ground truth, but practitioners often do not, which makes explicit support checks essential. Accuracy and AUC remain necessary, but for valid prevalence estimates across contexts they are not sufficient.

Data availability. Replication code is available at https://github.com/facebookresearch/multicalibrated_llm_measurement and will be deposited in the Political Analysis Dataverse on acceptance. ACS data is public via *folktables*; CAP data is public at <https://www.comparativeagendas.net>. Claude Opus inference outputs for all 30,000 main-text documents and Llama 3.3 70B outputs for all 105,000 documents used in SI Section S2 are included in the replication materials. A disclosure of generative-AI use in preparing this manuscript is provided in SI Section S8.

Competing interests. The authors declare no financial competing interests. In the interest of full disclosure, several authors are affiliated with Meta Platforms, and a subset are authors of the MCGrad algorithm used here; the open-source implementation is maintained by their team. The conceptual claim and the empirical guarantee are algorithm-agnostic, and any multi-accuracy or multicalibration fitter can be substituted for MCGrad. The open-weight Llama 3.3 70B replication (SI Section S2) shows the method does not depend on proprietary model access.

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